safeHavens and eSTZwritR; R packages for prioritizing germplasm collections for restoration and conservation

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Abstract

Introduction

Current estimates suggest that 45% of flowering plant species are at risk of extinction (Isbell et al., 2023; Bachman et al., 2024). Two growing responses to mitigate the loss of taxonomic and genetic diversity include habitat restoration for *in situ* conservation and the collection and storage of plant germplasm in *ex situ* collections (Volis and Blecher, 2010; Heywood, 2017; Westwood et al., 2021). Ecological restoration has become a tool to enhance habitat quality in natural areas or to convert land back to natural areas to provide connectivity between remnant parcels, helping to preserve plant and other biodiversity *in situ* (Gann et al., 2019). Long-term *ex situ* conservation is used when a species has, or is expected to experience, the loss or degradation of its habitat beneath thresholds that can support the species persistence in the wild (Oldfield and Newton, 2012; Mounce et al., 2017). However, analogously to *ex situ* conservation - albeit on a different time scale, the development of native plant propagules for ecological restoration requires the collection, and agricultural amplification of germplasm to meet the volumes required for use in restortaion projects [SHRIVER et al. 2026; Wambugu et al. (2023)].

Central to attaining germplasm collections is the acquisition of adequate amounts of genetic diversity for a portion of the species range for *in situ* restoration or of the entire species for *ex situ* conservation [Walters and Pence (2021); SHRIVER et al 2026]. Approaches for collecting germplasm from wild species for *in situ* restoration projects and *ex situ* in use by the nearly 1750 seed banks globally often follow a set of minimal rule-of-thumb guidelines developed from agricultural and population genetics studies (Hay and Probert, 2013; Westwood et al., 2021). Currently, guidance exists for conservation and *in situ* restoration projects, the number of propagules that can be collected to safeguard the persistence of natural populations (i.e., avoid over-harvesting), the minimum number of individuals that need to be sampled per population, and the number of populations to sample per species to reach genetic diversity targets (Basey et al., 2015; Nevill et al., 2018). AND guidance on the geographic structure of where to move restoration collections to and from to avoid using maladapted phenotypes at a restoration site (Vander Mijnsbrugge et al., 2010, CITE). However, a variety of goals exist at the institutional, programmatic, and bio-regional organizational levels, as well as opinions among curators, which have fostered tailored strategies beyond rule-of-thumb guidelines (Guerrant Jr et al., 2014). Furthermore, these minimal guidelines do not guarantee alignment or achievement of goals for all observed species and population genetic structures or population vital rates because of nuances in the biology and ecology of individual species (Whitlock et al., 2016; Hoban et al., 2020; Bucharova et al., 2025; Höfner et al., 2025).

However, the germplasm collections desired for ecological restoration and *ex situ* conservation differ in a key attribute. Restoration collections seek to maximize the likelihood of restoration success at sites in a defined focal area and time and seek to identify populations with the highest chance of doing so while maintaining some allelic diversity for at site selection (Johnson et al., 2010).

Ex situ germplasm collections, on the other hand, aim to maximize coverage of neutral genetic variation, the material which allows evolution a wide array of phenotypes to act upon, following the logic that no set of phenotypes are universally most fit, and the long term persistence of the species will require a wide base of standing diversity for selection to act upon.

Here, we present two geospatial software packages to help curators refine and implement their collection goals - using free and open-source software (FOSS) developed in the R programming language, and requiring minimal familiarity with R to use. The first tool, `safeHavens` (https://sagesteppe.github.io/safeHavens/), which is global in coverage, aims to maximize the putative coverage of neutral genetic variation across the entirety, or subsets of, a species range using proxies: 1) geographic, 2) ecoregional, and 3) environmental coverage of *ex situ* and restoration collections in areas lacking defined seed transfer zones, or in scenarios where seed transfer zones may not align with the goals of restoration collections. The second tool, tailored to the Western United States - but globally applicable, `eSTZwriter` (https://sagesteppe.github.io/eSTZwritR/) assists quantitative geneticists in producing standardized spatial data products describing areas for targeted seed collection and restoration planning, which may be readily used by curators.

(1)

[Figure 1 about here.]

safeHavens

While accessioning and maintaining germplasm collections *ex situ* to develop propagules for restoration and conservation activities is cost-effective, the acquisition of collections from the wild across a species range, especially from many populations and individuals within them, remains expensive due to costs associated with field work and staffing [Li and Pritchard (2009); CITE: field work is expensive- can we expert pers. comm Andrea or Kay on this?). Hence, the sampling of germplasm from a species aims to maximize the total genetic diversity across its range, or the suitability of populations for use in restorations, while minimizing the number of populations sampled. However, systems for guiding and measuring progress relative to this principle are absent for rare taxa, and for common taxa rely on data sets that may not relate to an individual species biology or ecology (e.g., the use of provisional STZs or ecoregions), or require considerable research to develop (eSTZs).

Critiques of provisional seed transfer zones, the most used method in the USA include (Durka et al., 2025,)... Limitations of alternatively proposed systems, such as sampling along ecoregions include (ask Chris and Dave, but also granularity).... Empirical Seed Transfer Zones, face limitations in their development as they require considerable empirical data, itself facing multiple challenges towards its collection, and given their cost are unlikely to be developed for the full range of species desired by restoration ecologists (cool guy from NV I like). Indeed, recent systems for selecting seed sources for specific restoration site have moved to developing computational methods for selecting material using existing data (St. Clair et al., 2022; Silva et al., 2025). However, these systems require that the restoration site be identified *before* treatments need to be applied, knowledge that exists for planned restoration projects or the development of material for a particular conservation unit, but that would require a lag of several years of germplasm material development for restorations after natural disturbances - especially across large geographic domains or reserves (Shriver et al. 2026).

Accordingly, we present `safeHavens` a landscape genetics inspired (?) package for R that helps curators develop taxon specific sampling schemes to maximize either the neutral genetic diversity of taxa using null models or have suitable candidate material for a restoration site available.

safeHavens partitions species ranges into geographically constrained groups by utilizing clustering of three types of distances: geographic, or resistance to gene flow, or environmental, each of which has been empirically shown to influence species genetic structures (Slatkin, 1993; McRae, 2006; Wang and Bradburd, 2014). These clusters, represented by spatial data, can be used to design germplasm collection endeavors, assigning a desired number of target samples to each cluster. The spatial data generated by this process can then be sent to field crews to guide both the planning and implementation of collections.

Usage

Functions that utilize geographic distance include: `PointBasedSample`, `EqualAreaSample` and `IBDBasedSample`, and `OpportunisticSample` which is derived from the `PointBasedSample` approach but designs additional sampling around existing collections. Rather than optimizing designs specifically for maximizing distances directly, they partition geographic space into relatively equally sized increments that collections should be made from within. These functions are all highly adaptable to support `n` collection areas, and a single collection per geographic unit. A general vignette on the package, which covers these functions is available at Getting Started.

Landscape resistance to geneflow is modelled using the function `IBRSurface` and its data preparation helpers: `buildResistanceSurface`, `populationResistance`, and optionally the final sample draw to be generated using `PolygonBasedSample`. A single surface can be used for a landscape, or surfaces can be parameterized for individual species. The isolation by resistance workflow creates a *theoretical* surface, populated with user specified values, that represents the cost to movement of genetic material (e.g. fruits and seeds, and pollen) between populations, see Zeller et al. (2012) for caveats. A vignette covering the preparation of surfaces is available at Isolation by Resistance.

Environmental distance, akin to the concept of isolation by environment, tailoredto the biology of individual species, is accommodated by `EnvironmentalBasedSample`, and its data preparation helpers: `elasticSDM`, `PostProcessSDM`, `RescaleRasters`, `writeSDMResults`. This workflow seeks to identify user supplied variables associated with species occurrences, and cluster the species range into n specific regions using these variables and hierarchical clustering approaches, optionally the surfaces can be resampled using `PolygonBasedSample`. A vignette is available for this workstream at Species Distribution Models.

`PolygonBasedSample` works on any type of polygon data, but is focused on supporting either provisional or empirical seed transfer zones, or ecoregion level data. `PolygonBasedSample` is able to incorporate multiple user specified decisions to maximize sample numbers in various strata based on their size, number of disconnected areas, or - when the data are available, temperature. Accordingly, it can loosely accommodate the variables that went into the generation of seed transfer zones or ecoregions. A visual guide to these function is available at Polygon Based Sample

Additionally, a function for sampling rare species based on occurrence data `KMedoidsBasedSample` is included, which can run on geographic, resistance, or environmental distances. When using geographic distances as input, and if coordinate uncertainty data available, this function will jitter site coordinates within the measures of uncertainty so simulate actual population positions, it will also 'drop' populations that may be extinct, incorrectly identified, or of another taxon to give results that are resilient to these types of data imperection. The function uses co-location network strength to identify a set of populations that minimizes the sum of distances in 'n' user specified groups. A vignette is available Rare Species

A final function, that exists for creating cartographic boundaries to for use with mobile GIS software is `PrioritizeSample`, which creates concentric rings around the center of each sampling unit to try and guide collectors towards center regions of collection zones rather than extremes. It is detailed only in a vignette that emulates the total process of designing an actual sample design for species in Worked Example.

Detailed Functionality

The isolation by resistance workflow is separated into three main phases to allow users to review, modify, and tailor intermediate objects. Resistance surface rasters are created by `buildResistanceSurface`, followed by the sampling of these surfaces to create distance matrices using `populationResistance`, and finally, the classification of the species range into clusters with less resistance to movement within than between them by `IBRSurface`, the polygons from `IBRSurface` are then submitted to the general sampling function `PolygonBasedSample`. A cost surface raster that represents resistance to gene flow for the species across the landscape, used for calculating least-cost distances between sites, is parameterized in `buildResistanceSurface` it natively supports layers for oceans, lakes, rivers, topographic ruggedness, and a species distribution model's probability surface, but also allows users to incorporate additional features and weights as desired.*** Pairwise distances across the cost surface and between randomly sampled sites across the buffered species range are used to create a distance matrix for clustering in `populationResistance`. Sites are hierarchically clustered into n groups, and pixels of their buffered species range are iteratively assigned to each cluster using `IBRSurface`, using NbClust with a silhouette method [NCLUST]. `IBRSurface` will return a polygon sf object of all clusters that work with `PolygonBasedSample` to assign a number of desired collections to each identified cluster.

The data preparation for the `EnvironmentalBasedSample` is also compartmentalized into several functions, allowing users to review, modify, and save intermediate outputs. A species distribution model, using user-specific independent variables (raster format) supplemented with internally derived low-order PCNM surfaces, with internal occurrence data thinning and pseudo-absence generation, spatial cross-validation folds, and automatic variable selection using elastic net regression and recursive feature elimination, is prepared using `elasticSDM` (Kuhn and Max, 2008; Aiello-Lammens et al., 2015; Naimi and Araujo, 2016; Tay et al., 2023; Meyer et al., 2025; Oksanen et al., 2025). The predictions of habitat suitability are converted from probabilities into a binary surface of presence/absence using a user-specified threshold metric, and areas of known occupancy are optionally 'burned' back onto this raster surface, the result being a surface that tries to maximize coverage of the species' occupied habitat, using `PostProcessSDM` (Hijmans et al., 2024; Hijmans, 2026). The rasters, used as independent variables, and with terms included in the final sdm model, are then rescaled, based on their contributions (beta-coefficients) to the model, for use in identifying clusters of the species range that experience more similar relevant habitat characteristics than other portions, using `RescaleRasters`. Results from these processes may then be saved using `writeSDMresults`. The germplasm collection sample is implemented by `EnvironmentalBasedSample`, which samples and clusters n locations across the raster stack, a k-nearest neighbor classifier is then trained on a relatively class-balanced resampling off these clusters, and spatial clusters are identified using predictions from it (Venables and Ripley, 2002). If a user desires more than one germplasm collection per cluster, they may use `PolygonBasedSample` with the output from `EnvironmentalBasedSample`.

`KMedoidsBasedSample` is intended for use with rare species and maximizes the geographic coverage (by minimizing the sum of dissimilarities) of n collections by identifying individual populations to sample, in consideration of common data quality issues in occurrence data. `KMedoidsBasedSample` is the only function in `safeHavens` that directly models raw population data rather than generalized polygonal geometries around them. From user-specified data, occurrence records that are to undergo coordinate jittering, and any that may undergo population dropout (due to either misidentifications, conflicting taxonomy, or local extinction), are identified. A user-specified number of bootstrap simulations permute eligible sites, and updated distance matrices are fed into K-medoids repeatedly, within each run multiple restarts are allowed, to minimize the cost of the solution for each iteration (Maechler et al., 2025). Across all bootstrap iterations, each unique combination of candidate sites (set) is counted to identify the network of sites with the lowest cost (highest co-location strength), and the individual sites within this network are ordered by the number of times they were returned across all bootstrap simulations.

eSTZwritR

Empirical seed transfer zones (eSTZs) are gaining popularity among restoration practitioners as tools to identify the most appropriate seed source for a species at a restoration site (McKay et al., 2005). eSTZs are becoming more widely used for two primary reasons: 1) they are based on empirical data, such as phenotypes in a common garden, and 2) there are generally fewer zones than provisional seed transfer zones, thereby reducing the number of lineages that require cultivation in agricultural settings. Although popular, the development of eSTZs for a species is a costly and time-consuming process, most often involving common gardens or genetic studies, with many populations from across the species range incorporated as samples (Kramer et al., 2015).

While practices for developing eSTZs are becoming more defined, to our knowledge, no standards exist for *sharing* eSTZ results (Supplemental 1). Although eSTZs are produced by a handful of research of groups, inconsistencies vary across the spatial data products used to report and share eSTZs.

The success of a restoration project relies on the timely application of techniques that are suitable for the site at hand. Hence, the dissemination of ideas during and after a restoration project is the best opportunity to improve restoration outcomes (Figure 1). However, spatial data are notoriously complex and difficult to transmit, historically requiring specialised software, and spatial data require both written and geographic data (e.g., coordinates and relations between them) in the form of spatial data products (e.g., rasters and shapefiles) to accurately convey their meaning. Given the relative complexity of communicating precise spatial information, standards should exist to ensure not only its accuracy and precision but also its ease of interpretation and use.

Software

To make these suggestions easy to implement we have created an R package, eSTZwritR (pronounced 'easy rider'), which can implement all of them, less the statistical processing, with minimal user inputs. The package is installable from GitHub https://github.com/sagesteppe/eSTZwritR and has a GitHub Pages website (https://sagesteppe.github.io/eSTZwritR/) for users interested in better understanding its functionality, and which includes supplemental figures and details not discussed here.

The package requires only five functions to produce a directory with the contents discussed above, with minimal data entry. Most importantly, the entries are well outlined and easily entered without requiring close attention to detail. These results should allow for simple utilization of existing empirical seed-transfer zone resources. This software was designed based on a review of the data sets below…

Methods

Using 23 sets of eSTZs produced for 22 taxa, across the western United States, we showed that most of the spatial data developed and disseminated to share the results of an eSTZ are inconsistent (Supplemental 1). We have already observed significant hindrances to the uptake of these data at the practitioner level and searched for consensus within these data. Subsequently, using any consensus (wisdom of the masses) from these data combined with standard conventions of data sharing, we present a set of guiding standards for researchers in the United States to make the results more consistent.

We conducted a review of all eSTZs on the Western Wildland Environmental Threat Assessment Center (WWETAC) website as of May 1, 2024 (https://research.fs.usda.gov/pnw/products/dataandtools/datasets/seed-zone-gis-data). Each data product (file name structure, field naming conventions, and directory structure) was manually scored, and all analyses were performed in R version 4.2.1.

Results

[Figure 2 about here.]

In Figures 2 through 4, we present inconsistencies that we believe, or have observed, to interfere with practitioners' workflows. We encountered considerable inconsistencies within file names (Figure 2), in directory structure and naming (Figure 3), and cartographic elements of the 20 maps available (Figure 4). While some consensus existed around the use of USDA NRCS-Plants codes for denoting the taxon contained in the file (Figure 2), the lack of file names mentioning what attribute about the taxon they contained (e.g. 'zones,' 'seed_zone,' 'sz'), and the lack of specified geographic extents can make determining the specifics of the file difficult unless it is explicitly opened in a Geographic Information System (GIS) software.

[Figure 3 about here.]

The naming of the fields (columns) within vector data file presented the most problematic results (Figure 3), here we focus only on three inconsistences. Different uses of polygon geometry were implemented to represent the individual seed transfer zones, that is, sometimes all portions of a seed transfer zone, when at least some components are disconnected, where stored within the same object or row (a multipolygon). At other times, each discontinuous portion of the range was stored as its own polygon. For most infrequent Geographic Information System (GIS) users, we observed that multipolygons can be confusing and require them to use several moderately advanced spatial techniques to interact with. Surprisingly, within each shapefile, the field denoting the seed zones was often ambiguously labelled or entirely lacking any indication (Figure 3). In several instances, it took several minutes to determine which field was the seed zone by toggling through and visualizing many fields, despite us having already interfaced with all of these products multiple times.

[Figure 4 about here.]

Discussion

Some consensus exists among eSTZ developers for a range of attributes related to the distribution of data products. Combining these opinions with best practices for data sharing and experience as users of each of the existing empirical products results in the following recommendations.

Directory Structure

[Figure 5 about here.]

eSTZs should be distributed using a predictable directory structure that allows users to be immediately familiar with where to find the content (Figure 5). We recommend that all directories (folders) have two main subdirectories (Figure 5), one containing the essential data products, preferably in both raster and vector data formats (*see 'Data Formats'*). The second directory contains information related to the product, including a formatted citation for data use, a map for quick reference, and any materials describing the development of the product, both as a paper and a text file of quick metadata attributes.

File Naming

[Figure 6 about here.]

The files within the directory should follow an easily interpretable naming convention that allows users to import todata various softwares while also describing the essential attributes of the data product. We recommend (Figure 6) each file name has three main components. The first component is the USDA PLANTS code, and the second is the method used to develop the STZ - currently one of 'g', 'cg', 'cm' (for genetic, common garden, and climate matched, respectively), and the final is up to the two main regions which the product overlaps. In the United States, we recommend the use of the 12 Department of Interior regions as they cover contiguous geographic expanses, are few enough to be easily remembered, and balance L2

Omernik ecoregions with easier to remember state lines. However, in other nations, the use of ecoregions may be more desirable.

Maps

Maps of the data product should be included in the 'Information' directory. Many questions about eSTZs can be answered quickly and simply by a practitioner consulting a map saved as a PDF with the essential cartographic components; fortunately most developers already supply these. We recommend that each map contains the following elements: north arrow, scale bar, state borders, geographically relevant cities, coordinate reference system information, sensible categorical color schemes for the seed zones (e.g. from ColorBrewer https://colorbrewer2.org), a legend, the taxons name as a title, and the maps theme ('Seed Transfer Zones') as a subtitle.

Data Formats

We recommend that the spatial data associated with an eSTZ be distributed using both popular spatial data models, vector and raster. For vector data, we advocate the continued usage of the shapefile format, whereas for raster data, we propose the use of geoTIFFs ('tifs,' the. tif extension). In our experience, tifs seem to be the most widely used raster data models in ecology for non-time series data, and are supported by effectively all GIS software.

Vector Data Field Attributes

[Figure 7 about here.]

The order of the fields (or columns) of the vector data should follow a predictable pattern (Figure 7), allowing humans to interact with the data in a graphical user interface (GUI) to quickly detect their field of interest.

We recommend that each vector file have at least four fields in the following order and of the following data types. 1) The numeric integer (ID) a unique number associated with each individual polygon in the file. 2) Seed Zone (numeric integer) a unique identifier for each of the eSTZs delineated by the product developers, which allows for quick filtering of the data based on a simple numeric value that is difficult to misspecify. 3) SZName (character): a human-developed name for the zone that may refer to an axis of a principal component analysis, for example, 'LOW MEDIUM LOW', or be defined by the product developers. We propose that semi-informative names should be developed before data distribution to help practitioners more easily convey important attributes.

4) AreaAcres (numeric - integer) of each polygon.

In addition to these standard field naming and placement conventions, we recommend a series of standards for the contents within these essential fields and how to format any additional fields relevant to the project (see package website).

Discussion

Both packages are intended to increase the availability of native germplasm to increase the chances of success for terrestrial restoration and *ex situ* conservation. safeHavens supplies curators, and planners, with a set of tools to clearly determine options for specific sampling programs, which may be used for evaluating nad reporting progress towards conservation goals, as well as guiding field sampling activities. eSTZwritR helps research groups communicate and share the results of their findings wih germplasm curators and these curators with field crews, and restoration ecologists with loosely determing seed lots although tools such as *Seed Lot Selection tools* and AWESOME SA PAPER are now preferrable for most of these applications.

Both packages ensure that curators and groups do not require traditionally expensive Geographic Information System software, and are able to be readily modified and extended for various goals and utilities.

Author Contributions

R.C.B. conceived of the ideas, developed the software, and wrote the manuscript. B.W. contributed to the conceptualization of the eSTZwritR package, and collected all data for analysis of existing eSTZ products, and contributed to writing the manuscript.

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FIGURES

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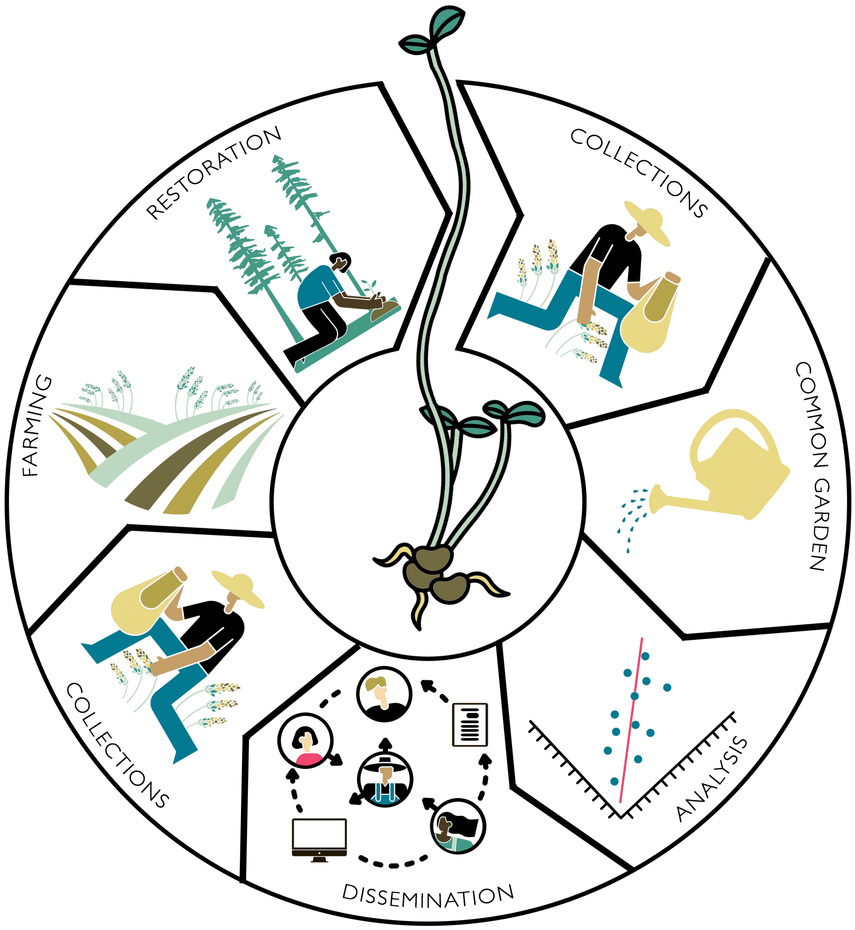

Figure 1: Dissemination, proceeding clockwise from 'Collections'. The first phase of collections includes the initial collection of germplasm from the wild, the most common starting point for both restoration and *ex situ* conservation. The next two panels *largely* refer to quantitative genetics approaches for the development of empirical seed transfer zones, while 'Dissemination' depicts the essential component for developing a curated collection of being able to convey priority areas to collector crews for a more targeted collection. These activities lead to the potential for agricultural or propagation increase of collected germplasm and the selection of materials for restoration plantings. By Emily Woodworth.

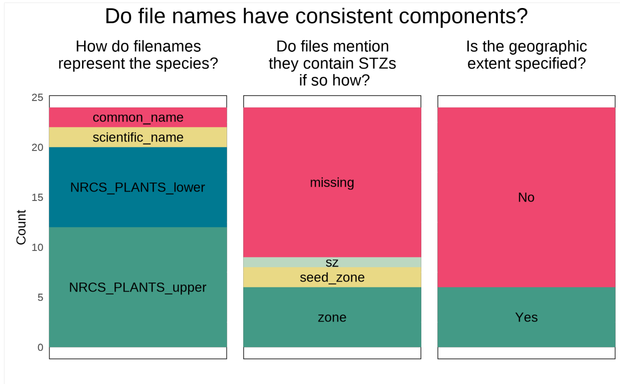

Figure 2: File Naming. Three inconsistencies in file names discussed here, with the advised format for data sharing in green, and the least desirable condition in grey.

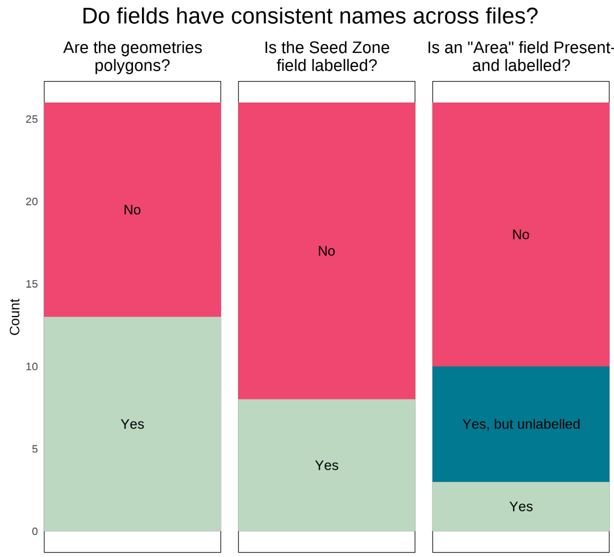

Figure 3: Field Names Shapefile. The three attributes of field names discussed here, with the most desirable condition in green, and the least desirable condition in grey.

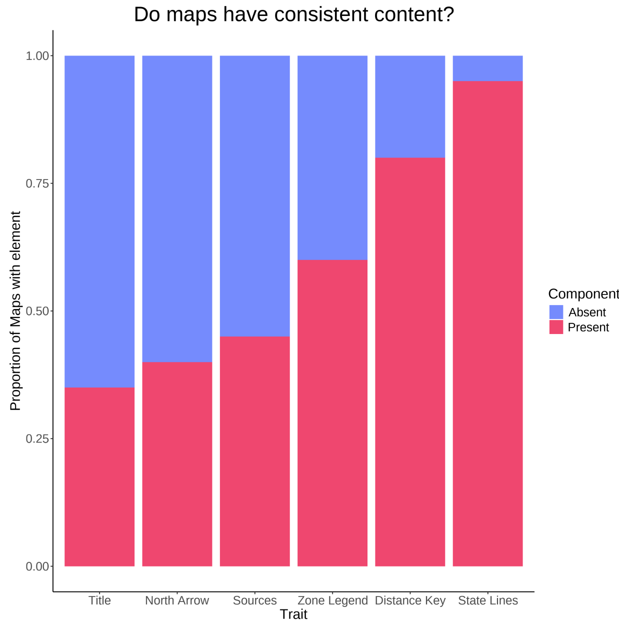

Figure 4: Map Components (n = 20). Several essential cartographic elements - most notably a Title, a statement on data sources, and a legend for the seed zones, where missing from at least - or nearly half of the products inspected.

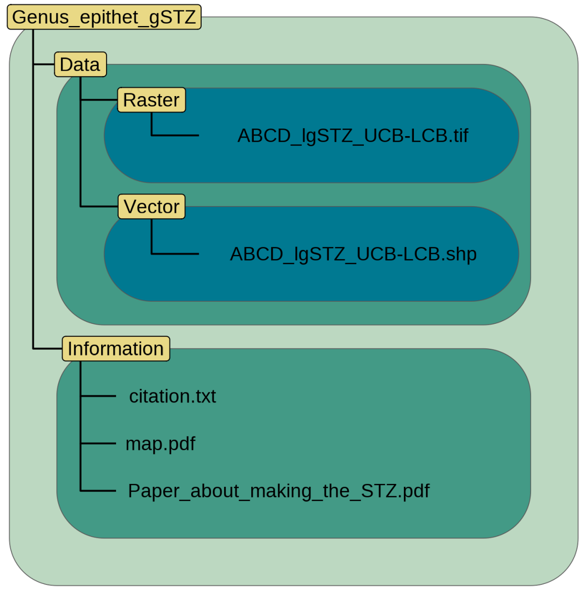

Figure 5: Directory Structure. Each directory is named in yellow, and spans the extent of variously coloured polygons. Individual files (or a set of files in the case of a shapefile) are depicted in black text within these directories.

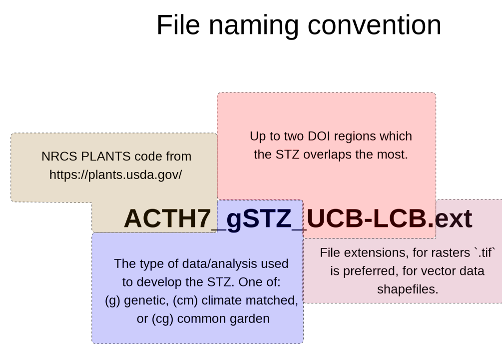

Figure 6: File Naming. The four proposed components of a filename are highlighted in different colours, and with appropriate cases.

Example field names in a shapefile					
ID	SeedZone	SZName	AreaAcres	BIO1_R	BIO2_mean
1	1	Salt Desert	12340	20.2	5.1
2	2	Desert Scrub	14230	19.1	7.1
3	3	Pinyon-Juniper/Oak Brush	30142	15.1	10.1
4	4	Montane	9872	12.3	12.3
The first four (blue) fields should be in every file. More fields are optional.					

Figure 7: Vector Data Field Attributes. The proposed field names for distributing vector data.